MUHAMMAD WALEED KHAN

Senior Frontend Engineer / Tech Lead — React, Vue.js, TypeScript, Node.js, Micro-Frontends

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SUMMARY

Senior Frontend Engineer with 9+ years shipping large-scale React and Vue.js applications for US retail and healthcare. Deep in micro-frontend and mono-repo architecture, design systems, frontend performance, and testing strategy. Has led frontend teams since 2021 — now Tech Lead at Nisum, still hands-on in architecture and code review.

CORE SKILLS

Languages: JavaScript (ES6+), TypeScript, HTML5, CSS3, SCSS

Frameworks: React, Vue.js 2/3, Node.js, Express, Redux, Redux Thunk, Vuex, Pinia, React Router, RxJS, Context API

Architecture: Micro-frontend (Module Federation), mono-repo, Backend-for-Frontend (BFF), design systems, component libraries, design patterns, SOLID, code splitting

AI-assisted development: GitHub Copilot, OpenAI Codex, Claude Code, Cursor; prompt engineering, AI-assisted PR review, test generation, refactoring and tech-debt cleanup, agent-driven bug triage

Performance & quality: Core Web Vitals (LCP, CLS, INP), Lighthouse, bundle analysis, lazy loading, caching, cross-browser support, WCAG 2.1 AA accessibility, semantic HTML, ARIA, responsive design

Testing: Jest, Cypress, Playwright, Vue Test Utils, React Testing Library, unit / integration / E2E

APIs & data: REST, GraphQL, MongoDB, Firebase, Adobe EDS endpoints

Styling: Tailwind CSS, Material UI, Ant Design, Bootstrap, CSS Modules

Tooling: Git, Webpack, Vite, Yarn, NPM, CI/CD, GitHub Actions, Agile/Scrum, code review

Leadership: Technical estimation, solution design, delivery playbooks, capacity planning, client and stakeholder communication, multi-project ownership, mentorship

EXPERIENCE

Nisum — Karachi, Pakistan — May 2021 – Present

Global software consultancy; delivered frontend platforms for a Fortune-500 US home & furnishings retailer and internal products.

Tech Lead — May 2026 to Present (formalizing team leadership held in practice since 2021; promoted from Principal Software Engineer)

- Own technical estimation, solution design, and capacity planning across 3 concurrent projects and 12 engineers; primary technical contact for client PMs, including negotiating scope and pushing back on unrealistic dates.
- Documented frontend conventions shared across teams — when to split a micro-frontend, how shared components are versioned, TypeScript rules, and accessibility and performance checks required before a story is closed.
- Introduced GitHub Copilot and Claude Code into the team's daily workflow for boilerplate, test scaffolding, and first-pass PR review, with clear rules on what still requires human review — reducing time spent on repetitive work.
- Remain hands-on in design and code review, arbitrating tradeoffs (Module Federation vs. build-time composition, shared state vs. isolated stores) and documenting decisions for stakeholders.

Principal Software Engineer — May 2025 to Apr 2026

- Led frontend architecture and technical design across multiple concurrent client projects as the primary technical contact for cross-functional teams.
- Drove adoption of shared design patterns and reusable components, removing duplicated implementations of the same UI across 4 codebases.
- Added Lighthouse and bundle-size checks to the CI pipeline so performance regressions on key pages were caught before release.
- Led incremental migration of legacy Vue 2 modules to Vue 3 and TypeScript, using AI-assisted refactoring to speed up the mechanical changes while every PR went through standard review and testing.
- Provided architectural guidance and hands-on mentorship to 3 engineering teams, reducing rework and review cycles on concurrent projects.

Senior Software Engineer, Frontend — May 2021 to Apr 2025

- Architected the mono-repo and micro-frontend (Module Federation) foundation for ShopByStyle, a fixed-bid e-commerce program — runtime composition over a single SPA let independent teams deploy without coordinated releases, keeping a fixed-scope contract on schedule and its margin intact.
- Built a reusable Vue.js component library adopted across 5 applications and 40+ screens, cutting feature development time by 25%.
- Designed a Backend-for-Frontend layer over shared EDS endpoints, collapsing 6 client-side calls into one response contract and unblocking native mobile reuse.
- Led 4 developers and 2 QA engineers — set coding standards, review discipline, and test coverage, and owned production releases and post-release triage for the platform.

Pcybil — Remote, USA — Nov 2020 – Apr 2021

Software Engineer (MERN)

- Sole engineer on the company's first MVP: React frontend, Node/Express REST APIs, and MongoDB schema, delivered end to end.
- Built dynamic, input-driven API endpoints that replaced over-fetching with targeted queries, improving data-retrieval performance across the app.
- Reduced operational cost 15% by consolidating API calls and optimizing backend architecture to lower server load.

Diya Interactive — Karachi, Pakistan — Oct 2019 – Oct 2020

Software Engineer (JavaScript)

- Owned frontend for Nextu (100,000+ users globally) and Saaya Health, building responsive, cross-browser interfaces in React, Redux, and CSS3.
- Partnered with UI/UX on flow redesigns that lifted conversion rate 20%, then instrumented and analyzed in-app UI changes to guide the next iteration.
- Cut technical debt 30% in one quarter through systematic code review and refactoring, improving release stability.
- Mentored 2 junior developers on JavaScript, Redux, and documentation practice.

Earlier

Frontend Developer, Freelance — Remote, Pakistan (Dec 2018 – Apr 2019). Shipped AssemblyF, a founder/investor community platform (React, Firebase Cloud Functions), and a law-firm reporting dashboard; cut bug-fix time 30% with modular components and disciplined Redux/Thunk state.

UI Developer, Mictronix — Karachi, Pakistan (Jun 2017 – Oct 2018). Converted high-fidelity designs into pixel-perfect responsive pages with hand-written, framework-free CSS; rebuilt the client web app in native JavaScript and reduced reported frontend bugs 15%.

PROJECTS

PIP Micro-Frontend Platform · Nisum · Vue 3, Module Federation, Webpack, CI/CD. Split a monolithic product-information experience into independently deployable remotes with a shared host shell, letting 4 teams release on their own cadence.

One Design Journey · Nisum · Design system, Storybook, TypeScript. Unified design language and reusable components across products; aligned design and engineering on one token set and component API, cutting UI drift between teams.

Flexible Template System · Nisum · Config-driven rendering, JSON schema. Configurable page-assembly system that let non-engineering teams launch new pages and products without custom development.

LEADERSHIP & COMMUNITY

- Mentored 10+ engineers across teams, 3 of whom were promoted; own growth plans and performance feedback for direct reports.
- Interview and assess frontend candidates for client teams, and run internal sessions on micro-frontend architecture and testing strategy.

EDUCATION

BS, Computer Science — Sir Syed University of Engineering & Technology, Karachi, Pakistan **2012 – 2016**